Close the quarter for a private equity fund

Ritamm · Software Engineering Intern · Take-home exercise

WINDOW	EXPECTED EFFORT	SUBMIT	ALLOWED
Sunday 1 November 2026 10:00–18:00 IST (8 hours)	5–6 hours	Zip of repository and a short screen recording, once, through your personal Form link	AI tools, Excel, any Python libraries

If we agreed a different window with you, your release email gives your times. Please read this brief once end to end before you write code. Everything you need is here or in the data pack. Every company, fund, bank and person in this exercise is fictional. AI tools are allowed and encouraged; you must understand and be able to explain every line you submit.

Tech stack: Python 3.11 or newer. You may use any libraries you like.

1. The story

The Greatness PE Fund, L.P. ("GPF") is a private equity fund. Ten investors have committed USD 250 million to it, and it owns stakes in eight private companies. Every quarter, the fund's administrator must work out what the fund is worth: its **net asset value (NAV)**.

It is now early January 2023. You are on Ritamm's team, and we are closing the quarter ending **31 December 2022** (Q4) for GPF. You have three things:

1. The fund's **bank statement** for Q4, as a PDF.
2. **Last quarter's numbers** (30 September 2022): the valuation report and the fund's statement of net assets, as Excel files.
3. A data file `agent_b_data.json` from **The Cayman Valuation Partners** ("CVP"), the independent firm that values the fund's companies. It holds CVP's Q4 marks. Your Agent B holds this file and shares the marks with Agent A only over A2A.

Your job is to build **two agents that talk to each other over A2A**, turning these inputs into a checked Q4 NAV, and to explain your choices. This is the kind of work Ritamm automates for real funds.

2. Terms you will need

Term	Meaning in this exercise
GP and LP	The general partner (GP) manages the fund. Limited partners (LPs) are the investors. GPF has ten LPs, coded LP01–LP10.
Commitment	The amount an LP has promised to invest over the fund's life. It is paid in parts, when the GP asks.
Capital call	The GP asks LPs to pay in a share of their commitment. Money arrives in the bank as wires from LPs. An LP may pay in more than one wire, or late.
Distribution	Cash the fund pays back to LPs, usually after selling a company.
Management fee	The fee the fund pays its manager (the GP's adviser) each quarter. An expense of the fund.
Portfolio company	A company the fund owns a stake in.
Cost	What the fund paid for a stake, in USD.
Fair value, or "mark"	What a stake is worth at the valuation date, as set by the valuation firm. Stated in the company's own currency.

Valuation method	How the mark was set, for example market multiples, discounted cash flow or a recent transaction.
Unrealized gain	Fair value minus cost, for a stake the fund still owns.
Realized gain	Sale proceeds minus cost, for a stake the fund sold.
Accrued expense (accrual)	An expense the fund owes at quarter-end but has not paid yet. It is a liability.
NAV	Net asset value: everything the fund owns minus everything it owes.
NAV bridge	A walk from last quarter's NAV to this quarter's NAV that explains every change. It must add up exactly.
ECB reference rate	The daily euro exchange rates published by the European Central Bank. CVP uses them to convert EUR values to USD.
A2A	Agent2Agent, an open protocol that lets AI agents from different organizations talk to each other over HTTP using JSON-RPC 2.0. See section 5.

3. Your inputs

Input	What it contains
`GPF_Bank_Statement_Q4_2022.pdf`	Every cash movement in the fund's USD bank account from 1 October to 31 December 2022: date, description, reference, debit, credit and running balance. Descriptions are bank-style text and can wrap onto several lines.
`CVP_Valuation_Report_Q3_2022.xlsx`	CVP's marks at 30 September 2022 for each company: currency, cost, fair value, FX rate, fair value in USD and valuation method.
`GPF_NAV_Summary_Q3_2022.xlsx`	Sheet 1: the fund's statement of net assets at 30 September 2022 (cash, investments, liabilities, NAV), cost and fair value per company in USD, and LP commitments. Sheet 2: accrued expenses at 31 December 2022.
`agent_b_data.json`	CVP's Q4 marks, notes, and the fund's Q4 purchases and sales of companies. This is the data that Agent B (your Valuation & NAV Agent) owns. Agent A does not have it.

4. What to build

Build **two agents** that communicate over A2A, plus one command that runs them. Keep each agent separate, with a clear input and output, so that each could later be reused for other funds.

- **Agent A - Bank & Ingestion Agent.** Owns the bank statement PDF and the categorization of transactions. Exposes an A2A endpoint. Parses the statement, categorizes every transaction, and derives the fund's Q4 purchases and sales of companies.
- **Agent B - Valuation & NAV Agent.** Owns `agent_b_data.json` (CVP's Q4 marks), the Q3 numbers, and all downstream work: reviewing marks for anomalies, and computing the Q4 NAV and the NAV bridge. Exposes an A2A endpoint. Releases Q4 marks only after Agent A proves its purchases and sales match Agent B's records.

Agent A asks Agent B for marks; Agent B replies with a question (the fund's Q4 purchases and sales); Agent A answers from its own output; Agent B returns the marks. This is a two-turn A2A conversation.

Agent A - Bank & Ingestion Agent

- Read every transaction from the PDF: date, full description, reference, debit, credit and balance.
- **Must use an LLM** to put each transaction in one of these categories: `capital_call`, `distribution`, `management_fee`, `investment_purchase`, `sale_proceeds`, `dividend_income`, `interest_income`, `bank_charges`, `expense`, `expense_refund`, `accrual_settlement`. For capital calls and distributions, also extract the LP code; for purchases and sales, the company name. Free tiers are enough (see the appendix).

- Check the work in code: opening balance plus all movements must equal the closing balance, and each running balance must agree.

Agent B - Valuation & NAV Agent (internal steps)

- **Review:** compare each Q4 mark with Q3 and flag anomalies using the rules in section 6. Give a one-line reason for each flag.
- **NAV:** compute the Q4 NAV and the NAV bridge in section 7. The bridge must balance to the cent.

One command (required)

Your submission must be runnable with a single command from the repository root, with no manual steps. Example: `python main.py ./pack`. Inputs are read from the pack directory passed as the first argument (default `./pack` if omitted); results are written to `./output`. Document any extra arguments in your README.

- Put `main.py` at the top level of your zip, not inside a folder.
- We run your code on our machine with our own `LLM_API_KEY`, `LLM_BASE_URL` and `LLM_MODEL` (an OpenAI-compatible endpoint). Read all three from the environment; do not hard-code a provider or a key.
- Your agents read their ports from `AGENT_A_PORT` and `AGENT_B_PORT` (with defaults of your choice) and stop when the run ends.
- We delete the `output/` folder in your zip, run your code again and compare the two, so include the output from your own run.

5. A2A between your two agents

Both agents follow the A2A protocol, version 0.3, using JSON-RPC 2.0 over HTTP.

Item	Value
Agent card (public)	`GET http://<host>:<port>/.well-known/agent-card.json` for each agent
Endpoint	The `url` field in the agent card. Send `POST` requests there.
Method	`message/send` only.

The conversation

1. **Agent A asks for marks.** Sends a message with a data part `{"fund": "GPF", "valuation_date": "2022-12-31"}`.
2. **Agent B asks a question.** Replies with a Task in state `input-required`. The text part says what it needs: the fund's Q4 purchases and sales of companies. The task has an `id` and a `contextId`.
3. **Agent A answers.** Sends a second message with the same `taskId` and `contextId`, and a data part with `purchases` and `sales` (shape below). Send an empty list if there were none.
4. **Agent A gets the marks.** If the lists match Agent B's records, the Task is `completed` and `artifacts[0]` holds the marks and notes. If not, it stays `input-required` and tells you how many items matched. Fix your lists and send again.

Shapes

Second-turn message (Agent A's answer):

```
{"kind": "message", "role": "user", "messageId": "<new uuid>",
 "taskId": "<task.id>", "contextId": "<task.contextId>",
 "parts": [{"kind": "data", "data": {
    "purchases": [{"company": "...", "date": "YYYY-MM-DD", "amount_usd": 0.0}],
    "sales":     [{"company": "...", "date": "YYYY-MM-DD", "proceeds_usd": 0.0}]}}]}
```

Completed task (abridged):

```
{"kind": "task", "id": "task-...", "contextId": "...", "status": {"state": "completed", ...},
 "artifacts": [{"artifactId": "marks-GPF-2022-12-31", "parts": [{"kind": "data", "data": {
    "fund": "The Greatness PE Fund, L.P.", "valuation_date": "2022-12-31",
    "marks": [{"company": "...", "currency": "USD", "fair_value": 0.0,
               "valuation_method": "...", "valuation_date": "2022-12-31"}],
    "notes": [{"company": "...", "note": "..."}]}}]}]}
```

6. Review rules

Rule	Flag when	Notes
R1	A mark changed materially from Q3	You choose the threshold (see hint 3).
R2	The valuation method changed from Q3	A method change can move a value for reasons other than performance, so a reviewer should look at it.
R3	The fund holds a company at 31 December, but CVP gave no Q4 mark	Read CVP's notes. See hint 2 for the value to use.
R4	CVP gave a mark for a company the fund does not hold at 31 December	Leave it out of the NAV and flag it.

Three situations below need a judgment call. Each has a hint. There is no single right assumption, but there is a well-reasoned one. Write each assumption in `ASSUMPTIONS.md`: what you assumed, why, and what would change if you were wrong.

> **Hint 1: one mark is in euros.** CVP states fair values in each company's own currency. The fund reports in USD, so convert using the **ECB euro reference rate** for the valuation date, as CVP did in Q3. 31 December 2022 was a Saturday: decide which date's rate applies. Find the rate yourself and cite the source. Keep cost in USD as given in the Q3 files; do not re-convert it.

> **Hint 2: a company with no mark yet.** If the fund bought a company late in the quarter, the valuer may not have valued it yet. Common practice is to carry a recent purchase at its **cost** (the price paid) until its first valuation. Do not leave it out of the NAV.

> **Hint 3: what counts as "material".** There is no threshold in this brief. Pick one and justify it. Many funds use a band of about 10–25% for a quarter. Think about which currency you measure the change in: a move in the EUR/USD rate is not a change in a company's value.

7. NAV and the NAV bridge

In plain words: the fund's **NAV** is what it owns minus what it owes at quarter end. The **bridge** is a step-by-step walk from last quarter's NAV to this quarter's NAV, where every step is a real cash or value change. If you add all the steps, you must land exactly on this quarter's NAV.

> **NAV at 31 Dec 2022** = cash at bank at 31 Dec + investments at fair value in USD at 31 Dec - liabilities at 31 Dec (the accrued expenses on sheet 2 of the NAV summary)

> **NAV bridge:** NAV at 30 Sep + capital called (received in Q4) - distributions paid + income (dividends and interest) - expenses + realized gain + change in unrealized gain = **NAV at 31 Dec**

Definitions for this exercise

- **Expenses** = management fee + other fees and expenses paid in Q4, net of refunds and including bank charges + the Q4 accrued expenses.
- **Not an expense:** a payment in Q4 that settles a liability already recorded at 30 September (an `accrual_settlement`). It reduces cash and the liability by the same amount, so the NAV does not change. Check the Q3 statement of net assets for these liabilities.
- Treat every other expense paid in Q4 as a Q4 expense, even if it covers a later period.
- **Realized gain** = sale proceeds - cost of the stake sold.
- **Unrealized gain** at a date = the sum over stakes held at that date of (fair value in USD - cost in USD). **Change in unrealized gain** = the 31 December total - the 30 September total. When a stake is sold, its unrealized gain at 30 September reverses here.
- If the bridge does not balance, a transaction is in the wrong category. Fix the cause, not the number.

8. What your code writes

Write these files to an `output/` folder. Use the field names shown; you may add fields.

File	Contents
`transactions.json`	A list of transactions: `date` (YYYY-MM-DD), `description`, `reference`, `debit`, `credit`, `balance`, `category`, `lp_code` (or null), `company` (or null).
`a2a_log.jsonl`	One line per request and per response between your two agents, with a timestamp.
`marks.json`	The artifact data Agent A received from Agent B, unchanged.
`review.json`	One entry per company: `company`, `currency`, `q3_fair_value`, `q4_fair_value`, `change_pct`, `q3_method`, `q4_method`, `flags` (list of rule IDs, e.g. `["R1"]`), `reason`. Add `threshold_pct` and `eurusd_rate` at the top level.
`nav.json`	`closing_cash`, `investments_usd`, `liabilities`, `nav_q4`, and a `bridge` object: `nav_q3`, `capital_called`, `distributions`, `income`, `expenses`, `realized_gain`, `change_in_unrealized`, `nav_q4`, `difference` (should be 0.00).
`nav_workings.xlsx` (optional)	Your working file, if you used Excel to check your numbers.

Your repository also has:

- `README.md`: how to run it, how it is structured, what you would do next to take it from a prototype to a product, and what is missing or broken. Be honest about gaps.
- `ASSUMPTIONS.md`: your assumptions, including the three in section 6. A template (`ASSUMPTIONS_TEMPLATE.md`) is provided in the pack. Suggested format per assumption: `## Assumption N: ...` then `What I assumed`, `Why`, `What would change if wrong`.
- `DECISIONS.md` (optional but recommended): a short log of 3 decisions you made, each with the decision, the alternatives you considered, and why you chose it. A template (`DECISIONS_TEMPLATE.md`) is provided in the pack. Format: `## Decision 1: ...` then `**Decision:**`, `**Alternatives:**`, `**Why:**`.
- Tests for the parts you think matter most. Your tests must run offline, without calling an LLM or any other network service.
- No keys or secrets in the code. Read them from environment variables or config files. LLM credentials (API key and base URL) must come from a `.env` file or environment variables and must not be committed. A `.env.example` is provided in the pack listing the variables we set when we run your code.

9. How to submit

Submit through the **personal Form link** in your release email, by the end of your window. Sign in with a Google account to open it. **You can submit once**; later submissions are not counted, so check everything first. Keep the

repository private if you push it to GitHub; the zip is what we review.

The Form asks for:

1. Your candidate ID (already filled in) and your full name.
2. Your numbers, in USD with two decimals and no commas, copied from `output/nav.json`: closing cash, capital called, distributions paid, management fee, income, total expenses, realized gain, change in unrealized gain, investments at fair value, and NAV at 31 December 2022.
3. From `output/review.json`: the EUR/USD rate you used, your materiality threshold and the companies you flagged; from `output/transactions.json`: the number of transactions, not counting the opening balance line.
4. A **zip file** of your repository including the `output/` folder, under 10 MB.
5. A **screen recording of 60–90 seconds**, with your voice: run the one command on your own machine, then explain one decision you made. A phone video of your screen is fine. Under 100 MB.
6. The AI tools and LLM you used, the hours you spent, and a declaration that you did the work and submitted the Form yourself.

Make sure the numbers you type match your output files exactly.

Keep the submission lean. Exclude `__pycache__`, `.pytest_cache`, `node_modules`, `.venv`, `.git`, build artifacts and any large binaries. Submit only source, config, tests, the `output/` folder and docs. A lean zip is easier to review and avoids size limits.

10. Rules

- **AI tools and Excel are allowed and encouraged.** Tell us which you used. You must understand and be able to explain every line you submit.
- **Work alone.** Do not share code, answers, files. Copied work leads to disqualification of everyone involved. We compare submissions.
- **Questions:** email `hello@ritammfin.com` with the subject "Take-home question".

11. Planning your time

A suggested plan: 30 minutes to read and plan, 2 hours on the Bank Agent, 1 hour on the A2A pair, 1 hour on the Review and NAV Agents, 1 hour on tests, README and assumptions. If you run short of time, submit what works and say what is left. A partial, honest submission beats a broken complete one.

Good luck. We look forward to reading your work.

Appendix: LLM access

The Bank Agent must use an LLM. Any OpenAI-compatible provider works, and a free tier is enough for this exercise (about 40 transactions). For example, a free Groq key with the model `openai/gpt-oss-20b` and base URL `https://api.groq.com/openai/v1`, as in `.env.example`. More free options: github.com/mnfst/awesome-free-llm-apis. Rate limits change; check the provider's documentation.